

Neurons and Membrane Potential

A neuron is the basic signaling unit of the nervous system. Like other cells, it maintains a difference in electrical charge between the inside and the outside of its membrane. At rest, the inside of a neuron sits at about -70 millivolts relative to the outside, a state called the resting membrane potential.

This voltage difference exists because ion channels and pumps control the flow of charged particles across the membrane. The sodium-potassium pump moves three sodium ions out of the cell for every two potassium ions it moves in, using ATP as its energy source. Because the membrane is more permeable to potassium than to sodium at rest, potassium tends to leak outward, pulling the inside of the cell toward a more negative voltage.

When a neuron receives enough excitatory input, the membrane potential rises toward a threshold, usually around -55 millivolts. If that threshold is crossed, voltage-gated sodium channels open rapidly, sodium rushes into the cell, and the membrane potential spikes upward in an event called an action potential. This is an all-or-nothing event: either the threshold is crossed and a full spike fires, or it is not crossed and nothing happens, regardless of how strong the input was.

After the peak of the spike, voltage-gated potassium channels open and sodium channels inactivate, driving the membrane potential back down and briefly past the resting level, a period called hyperpolarization. During part of this recovery, called the refractory period, the neuron cannot fire again, which limits the maximum rate at which it can spike and gives each action potential a clean, discrete shape as it travels down the axon.

Action potentials propagate along the axon without losing amplitude, because each patch of membrane regenerates the spike rather than passively transmitting a fading signal. In myelinated axons, the wrapping of myelin around the axon forces the spike to jump between unmyelinated gaps called nodes of Ranvier, a process called saltatory conduction that speeds up transmission considerably compared to an unmyelinated axon of the same diameter.

Synaptic Transmission

When an action potential reaches the end of an axon, it arrives at a structure called the axon terminal, which sits close to another neuron's dendrite or cell body across a narrow gap called the synaptic cleft. Most communication across this gap is chemical rather than electrical.

The arrival of the spike opens voltage-gated calcium channels in the terminal. Calcium flowing into the terminal triggers vesicles filled with neurotransmitter to fuse with the presynaptic membrane and release their contents into the synaptic cleft, a process called exocytosis. The neurotransmitter molecules diffuse across the cleft and bind to receptors on the postsynaptic membrane.

Two broad receptor types exist. Ionotropic receptors are ligand-gated ion channels: when neurotransmitter binds, the channel opens directly and ions flow immediately, producing a fast postsynaptic response measured in milliseconds. Metabotropic receptors instead trigger a second-messenger cascade inside the postsynaptic cell, producing slower but longer-lasting effects, sometimes changing gene expression.

Whether a synapse is excitatory or inhibitory depends on which ions flow, not on the identity of the neurotransmitter alone. Glutamate binding to an ionotropic receptor typically opens channels that let sodium in, depolarizing the postsynaptic cell and producing an excitatory postsynaptic potential. GABA binding to its ionotropic receptor typically opens channels that let chloride in, hyperpolarizing the cell and producing an inhibitory postsynaptic potential. A single postsynaptic neuron integrates thousands of these excitatory and inhibitory inputs across its dendrites, and whether it fires depends on the net sum reaching threshold at the axon hillock.

After release, neurotransmitter is cleared from the cleft by reuptake transporters that pump it back into the presynaptic terminal, by enzymatic breakdown, or by diffusion away from the synapse. This clearance step resets the synapse for the next signal and is also the target of many psychoactive drugs, which slow or block reuptake to prolong the neurotransmitter's effect.

Hebbian Learning and Plasticity

Hebbian learning is often summarized as "cells that fire together wire together."

The idea, proposed by Donald Hebb in 1949, is that when a presynaptic neuron repeatedly contributes to firing a postsynaptic neuron, the connection between them strengthens, making future firing of the postsynaptic neuron by that same presynaptic input more likely.

The best-studied physiological form of this idea is long-term potentiation, or LTP. In a typical experiment, a brief high-frequency burst of stimulation to a presynaptic pathway leaves the synapse measurably stronger for hours or longer, seen as a bigger postsynaptic response to the same test stimulus. LTP in many pathways depends on the NMDA receptor, a glutamate receptor that is unusual because it requires both neurotransmitter binding and sufficient postsynaptic depolarization to open, since a magnesium ion normally blocks its channel at resting potential.

This dual requirement makes the NMDA receptor a coincidence detector: it only lets calcium into the postsynaptic cell when the presynaptic terminal releases glutamate at roughly the same time as the postsynaptic neuron is already active. That calcium influx triggers signaling cascades that insert additional AMPA receptors into the postsynaptic membrane, so the same amount of glutamate produces a larger response on the next signal, the physical basis of the strengthened connection.

The opposite process, long-term depression or LTD, weakens synapses when presynaptic activity occurs without matching postsynaptic activity, or when the pattern of coincident firing is different. Together, LTP and LTD let synaptic weights move in both directions, which is necessary for a network to both learn new associations and forget or revise old ones rather than only accumulating strength.

Because plasticity here is described at the level of individual synapses and depends on the precise timing between presynaptic and postsynaptic activity, this same coincidence-detection idea reappears in a more general form in spike-timing-dependent plasticity, and it inspired the class of artificial learning rules that adjust connection weights based on correlated activity.